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Assignment 3: Visualize the data using Python by plotting the graphs for assignment no. 1 and 2. Consider a suitable data set. Use Scatter plot, Bar plot, Box plot, Pie chart, Line Chart.

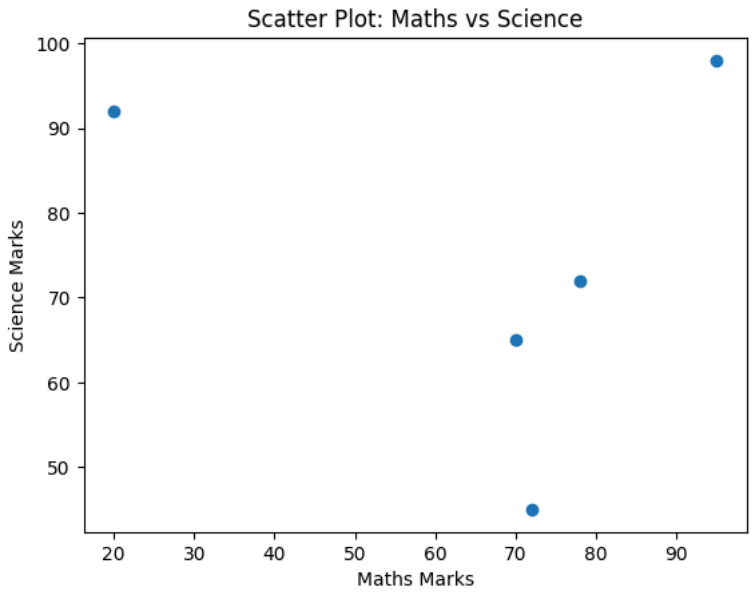
```
import pandas as pd
data = {
    'Student': ['Nisha', 'Sohaa', 'Apurv', 'Viraj', 'Vaishnavi'],
    'Maths': [78, 95, 72, 20, 70],
    'Science': [72, 98, 45, 92, 65],
    'English': [68, 80, 90, 85, 73]
}

df = pd.DataFrame(data)
print(df)
```

```
     Student  Maths  Science  English
0      Nisha     78       72       68
1      Sohaa     95       98       80
2      Apurv     72       45       90
3      Viraj     20       92       85
4  Vaishnavi     70       65       73
```

Scatter Plot

Purpose: Relationship between two numerical variables (Maths vs Science)

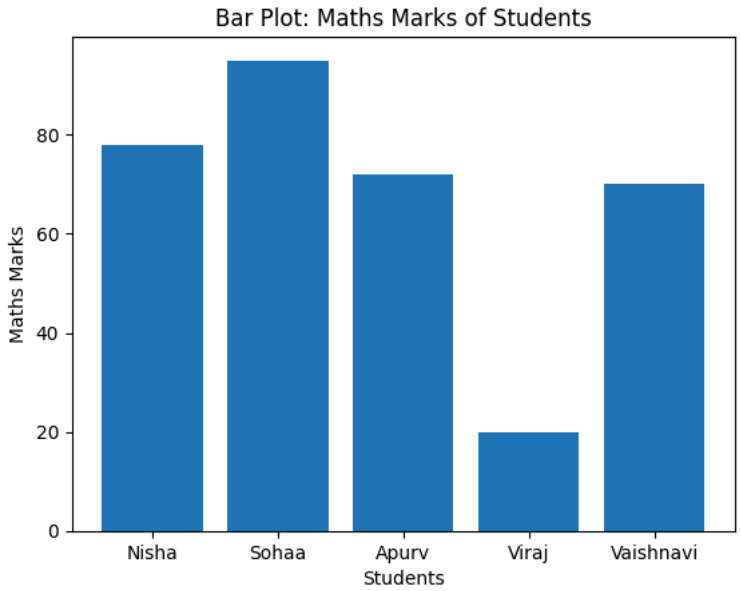
```
import matplotlib.pyplot as plt

plt.scatter(df['Maths'], df['Science'])
plt.xlabel('Maths Marks')
plt.ylabel('Science Marks')
plt.title('Scatter Plot: Maths vs Science')
plt.show()
```



Bar Plot

Purpose: Compare marks of students in Maths

```
plt.bar(df['Student'], df['Maths'])
plt.xlabel('Students')
plt.ylabel('Maths Marks')
plt.title('Bar Plot: Maths Marks of Students')
plt.show()
```

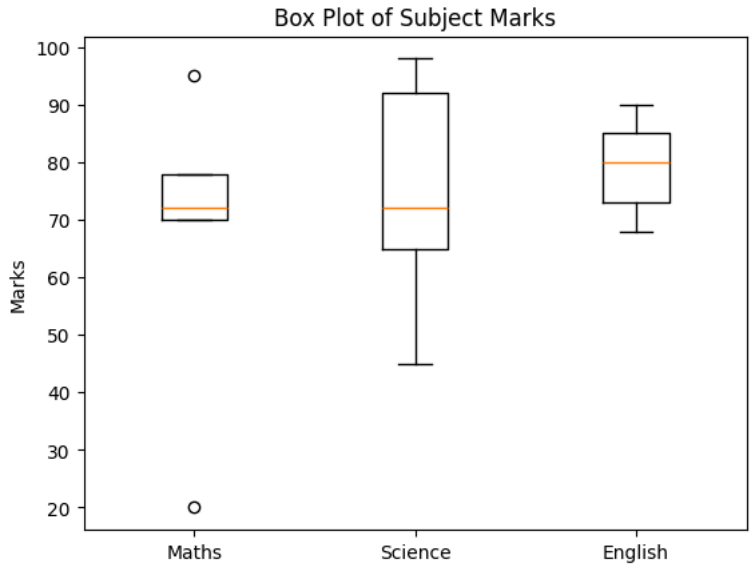


Box Plot

Purpose: Show distribution and outliers of marks

```
plt.boxplot([df['Maths'], df['Science'], df['English']],
            labels=['Maths', 'Science', 'English'])
plt.title('Box Plot of Subject Marks')
plt.ylabel('Marks')
plt.show()
```

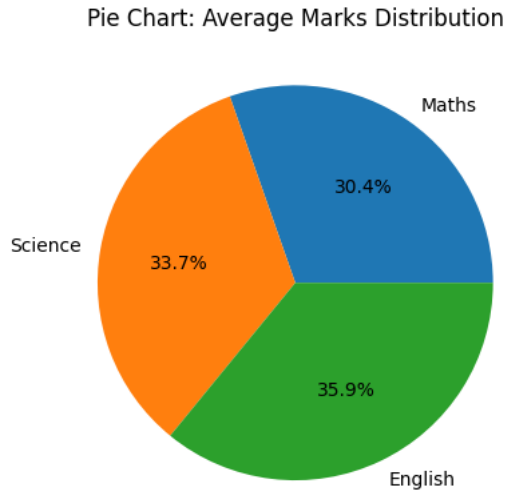
```
/tmp/ipython-input-2270569777.py:1: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick
  plt.boxplot([df['Maths'], df['Science'], df['English']],
```



Pie Chart

Purpose: Percentage contribution of total marks by subject (average)

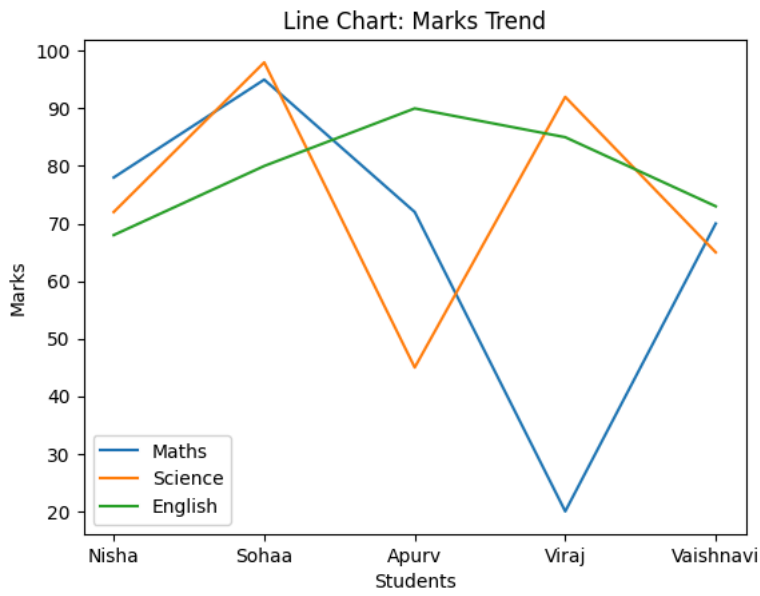
```
average_marks = df[['Maths', 'Science', 'English']].mean()

plt.pie(average_marks, labels=average_marks.index, autopct='%1.1f%%')
plt.title('Pie Chart: Average Marks Distribution')
plt.show()
```



Line Chart

Purpose: Trend of marks across students

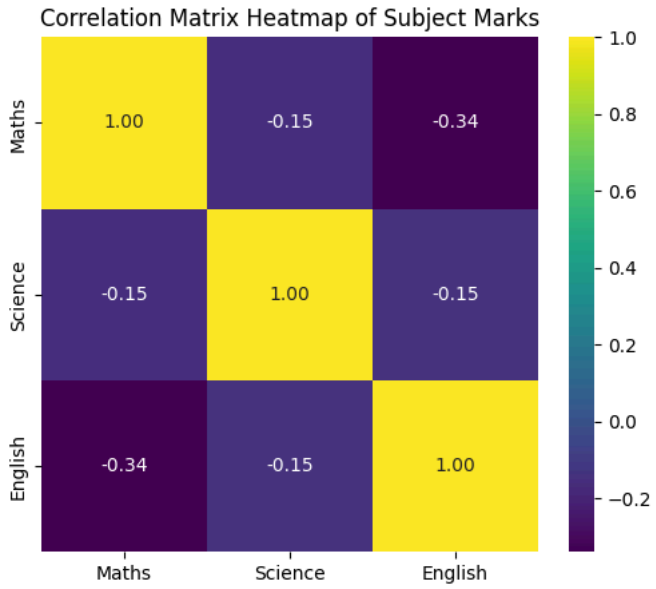
```
plt.plot(df['Student'], df['Maths'], label='Maths')
plt.plot(df['Student'], df['Science'], label='Science')
plt.plot(df['Student'], df['English'], label='English')

plt.xlabel('Students')
plt.ylabel('Marks')
plt.title('Line Chart: Marks Trend')
plt.legend()
plt.show()
```



Correlation Matrix - Heatmap

Purpose: Visualize the correlation between numerical features (Maths, Science, English).

```
import seaborn as sns
import matplotlib.pyplot as plt

correlation_matrix = df[['Maths', 'Science', 'English']].corr()

plt.figure(figsize=(6, 5))
sns.heatmap(correlation_matrix, annot=True, cmap='viridis', fmt=".2f")
plt.title('Correlation Matrix Heatmap of Subject Marks')
plt.show()
```


Correlation Matrix Heatmap of Subject Marks
Maths
Science
English
1.00
-0.15
-0.34
-0.15
1.00
-0.15
-0.34
-0.15
1.00
Maths
Science
English
1.0
0.8
0.6
0.4
0.2
0.0
−0.2